For Online Publication: Appendix A- Additional Figures and Tables

A1. Sample

Table A1.1—Referral Base for GWAS Participants by Diseases and Ancestries (per 100 Individuals)

Disease	White	Hispanics	Black	Asian	Native American
Allergic Reactions	2.52	1.22	2.10	na	na
Anemia and Other Deficiencies	0.81	1.00	1.08	na	na
Appendicitis	na	na	na	na	na
Cancer	3.33	0.49	0.96	na	na
Cerebrovascular Disease	0.85	na	na	na	na
Complications of Pregnancy and Birth	na	na	na	na	na
Congenital Anomalies	0.63	na	na	na	na
Copd, Asthma	18.28	11.49	13.89	10.66	19.98
Diabetes Mellitus	5.60	4.99	7.31	4.98	na
Disorders of Mouth and Esophagus	0.46	na	na	na	na
Disorders of Teeth and Jaws	2.94	1.78	2.67	na	na
Disorders of the Upper GI	5.91	2.98	3.64	na	na
Female Genital Disorders, and Contraception	6.59	2.80	3.40	na	na
Gallbladder, Pancreatic, and Liver Disease	1.33	0.56	na	na	na
Headache	1.87	1.01	0.88	na	na
Heart Disease	7.84	2.86	4.73	na	na
Hemorrhagic, Coagulation, and Disorders of White Blood Cells	na	na	na	na	na
Hereditary, Degenerative and Other Nervous System Disorders	0.96	na	na	na	na
Hyperlipidemia	11.63	4.21	6.00	7.15	na
Hypertension	16.57	8.57	17.96	11.03	14.35
Infectious Diseases	6.74	3.71	3.89	na	7.57
Intestinal Infection	2.49	2.89	2.43	na	na
Kidney Disease	2.28	1.00	0.98	na	na
Male Genital Disorders	1.73	na	na	na	na
Mental Disorders	14.32	7.28	8.18	3.15	14.77
Osteoarthritis and Other Non-Traumatic Joint Disorders	8.45	3.56	5.76	na	8.89
Other Bone and Musculoskeletal Disease	3.96	1.19	1.14	na	na
Other Circulatory Conditions and Arteries, Veins, and Lymphatics	3.09	1.15	1.81	na	na
Other CNS Disorders	5.84	2.11	3.10	na	na
Other Eye Disorders	3.12	0.99	1.47	na	na
Other Stomach and Intestinal Disorders	0.88	na	na	na	na
Otitis Media	3.97	3.43	2.18	na	na
Pneumonia	7.87	4.35	3.84	3.87	6.66
Skin Disorders	9.38	4.41	5.10	5.82	na
Systemic Lupus and Connective Tissues Disorders	5.76	2.81	3.93	na	na
Thyroid Disease	5.33	1.90	2.63	na	na

Notes: The table is created by the author based on the MEPS summary tables. Respondents in MEPS are classified into five ancestry groups: (1) Hispanic, (2) Black, (3) American Indian, Alaska Native, and multi-races, (4) Asian, Hawaiian and Pacific Islander, and (5) European whites. The minority population in our following analysis contains all the mentioned ancestries except the European whites.

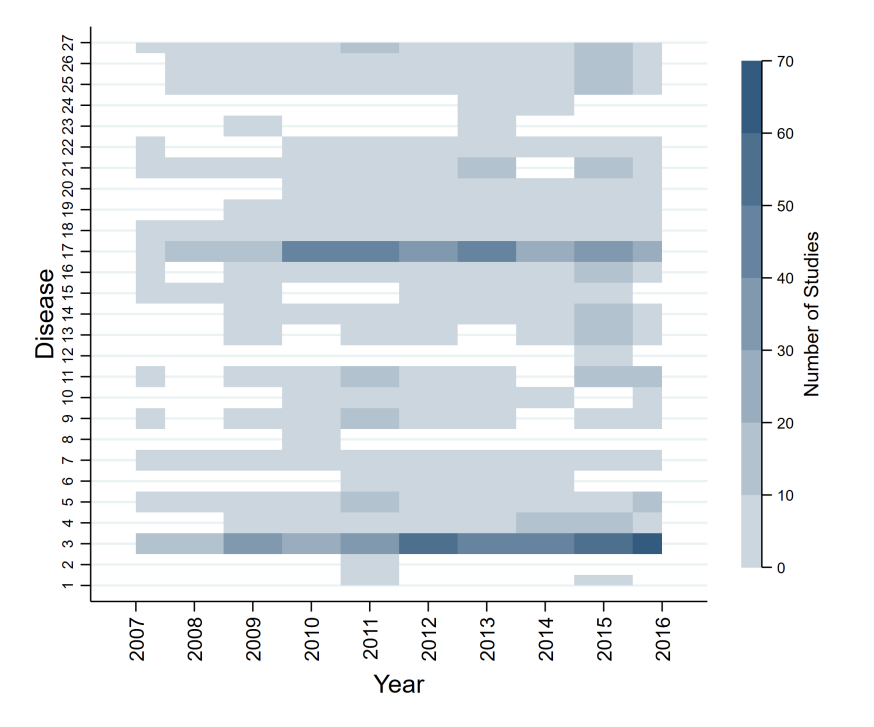
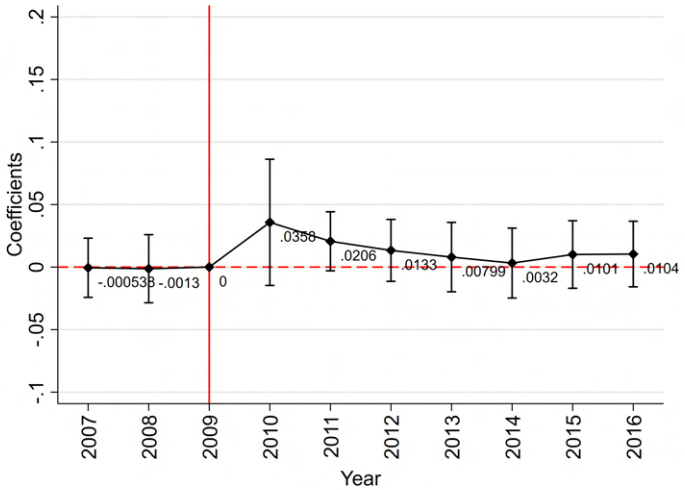


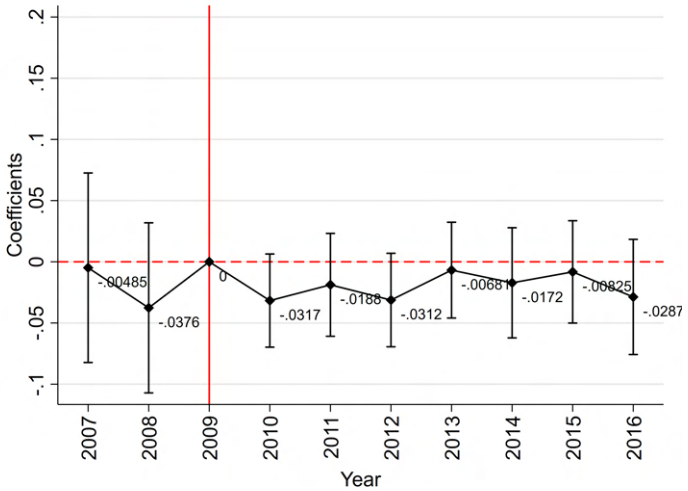
Figure A1.1. Heat Map for Disease Studies

Notes: The map sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. Given minimal GWAS publications in 2005 and 2006, the data in 2007 presented in the figure includes subjects from GWAS publications in 2005, 2006 and 2007. The IDs corresponds to the IDs in Table 1.

A2. ARRA-NIH Funding Approval Likelihood



(A) Share of Studies Supported by ARRA-NIH Grants over Studies Supported by Grants from Any Sources



(B) Share of Studies Supported by International Grants over Studies Supported by Grants from Any Sources

Figure A2.1. Impact on Grant Support

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. The sample is constructed based on studies supported by any U.S funding agencies between 2005 and 2016. The relative referral base is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of *RRB* and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.

Table A2.1—Impact on ARRA-NIH Grant Expansion on Grant Approval Likelihood

Dependent Var	Share of Studies Supported by ARRA-NIH Grants over Those Supported by US Grants		Share of Studies Supported by ARRA-NIH Grants over Those Supported by All Grants		Share of Studies Supported by International Grants	
	(1)	(2)	(3)	(4)	(5)	(6)
Relative Referral Base (2005) * Post	0.0150**	0.0181**	0.0130**	0.0165**	0.0006	-0.0088
	(0.008)	(0.008)	(0.007)	(0.007)	(0.016)	(0.018)
	[0.022]	[0.005]	[0.032]	[0.006]	[0.983]	[0.747]
	[0.0019, 0.0310]	[0.0056, 0.0336]	[0.0010, 0.0299]	[0.0045, 0.0330]	[-0.0485, 0.0546]	[-0.0598, 0.0453]
Disease Research Field Characteristics	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Disease FE	Yes	Yes	Yes	Yes	Yes	Yes
Outcome Mean (Pre-Policy)	0.041	0.041	0.034	0.034	0.210	0.210
Observations	183	183	183	183	183	183

Notes: The sample in column 1 comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. In column 2 and 3, we do not impose any restrictions on the source of funding. The dependent variable is specified in the column title. The relative referral base is the difference in the share of people with medical conditions between minorities and European whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. We report the Huber-White robust standard errors in the parentheses. We also report wild bootstrap cluster p-values in square brackets and wild bootstrap cluster 95% confidence intervals in square brackets, generated using boottest command (10000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at disease level with 27 disease clusters. Significance stars are labeled with respect to the robust standard errors. $*p < 0.1, **p < 0.05, ***p < 0.01$.

A3. Ancestral Diversity in GWAS Subjects

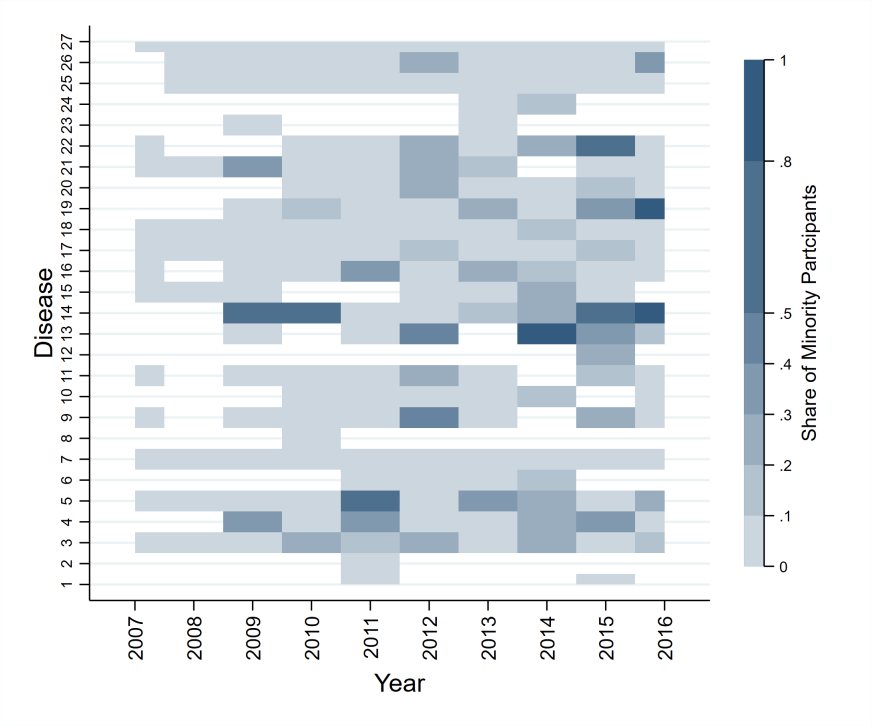


Figure A3.1. Heat Map for Ancestral Diversity in GWAS Subjects

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. Given minimal GWAS publications in 2005 and 2006, the data in 2007 presented in the figure includes subjects from GWAS publications in 2005, 2006 and 2007. The IDs corresponds to the IDs in Table 1.

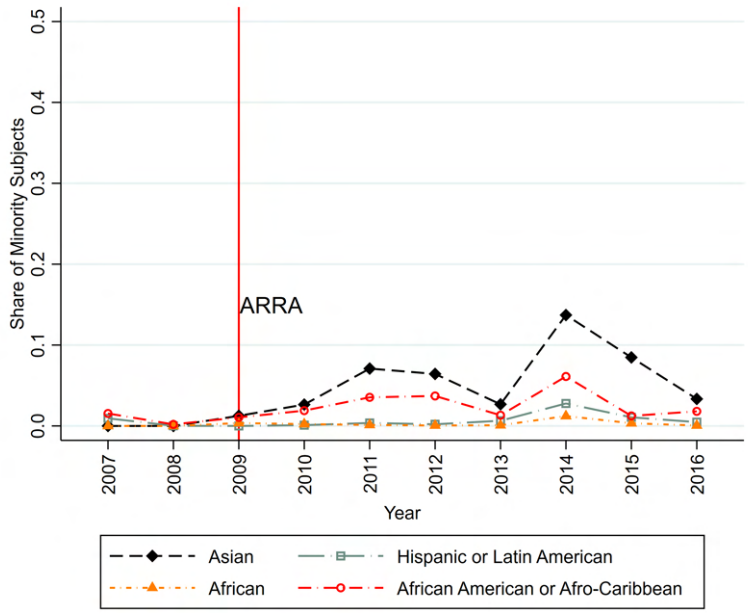


Figure A: Share of Minority Subjects in Discovery Sample by Races over Years

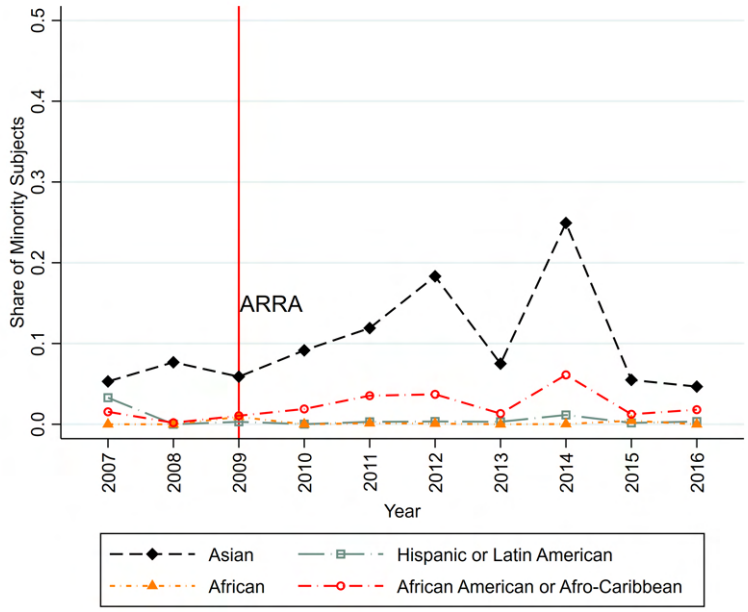
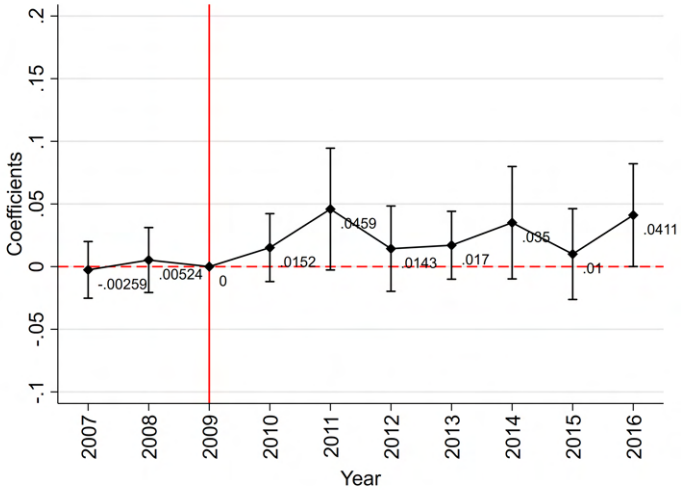


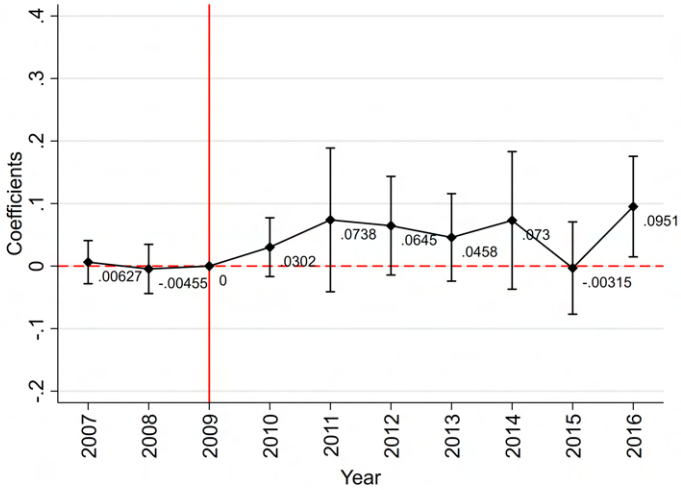
Figure B: Share of Minority Subjects in Replication Sample by Races over Years

Figure A3.2. Share of Minority Subjects by Races over Years

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 182 disease-year pairs for discovery stage analysis and 170 disease-year pairs for replication stage analysis. Given minimal GWAS publications in 2005 and 2006, the data in 2007 presented in the figure includes subjects from GWAS publications in 2005, 2006 and 2007.



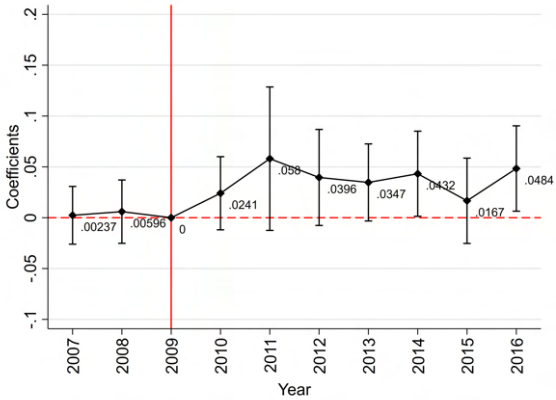
(A) Share of Minority Subjects in the Discovery Stage



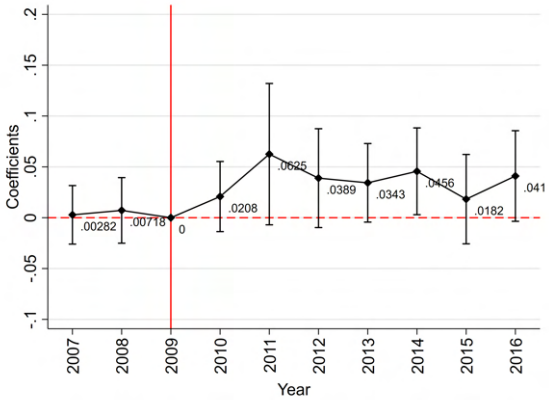
(B) Share of Minority Subjects in the Replication Stage

Figure A3.3. Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Subjects – By Stage

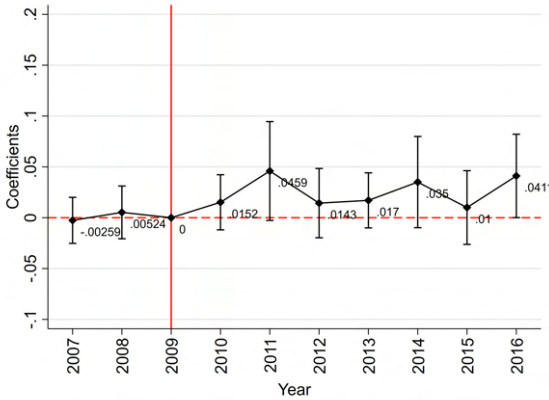
Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 182 disease-year pairs for discovery stage analysis and 170 disease-year pairs for replication stage analysis. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is the share of minority subjects for each disease in each year. The relative referral base (*RRB*) is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of *RRB* and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.



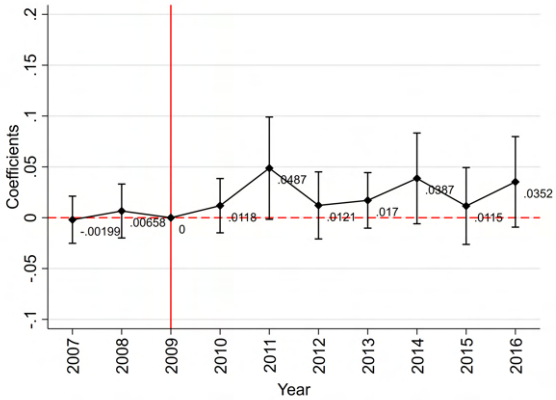
(A) RRB (2005), Discovery + Replication Stages



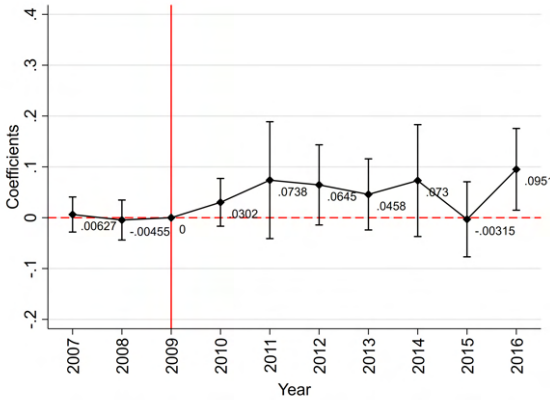
(B) Ave RRB (2002-2005), Discovery + Replication Stages



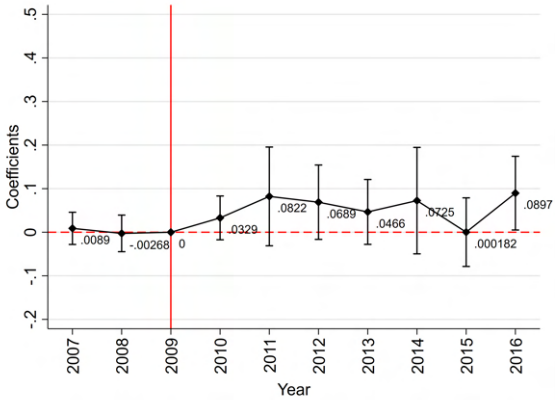
(C) RRB (2005), Discovery Stage



(D) Ave RRB (2002-2005), Discovery Stage



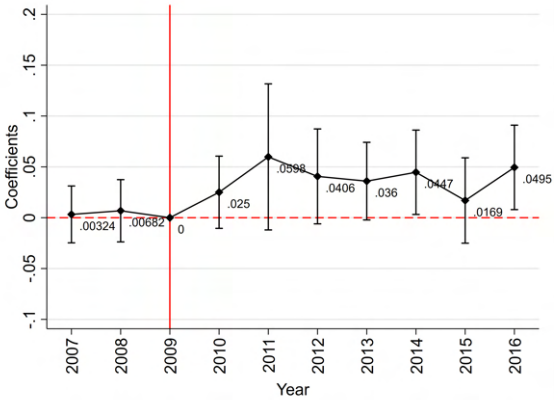
(E) RRB (2005), Replication Stage



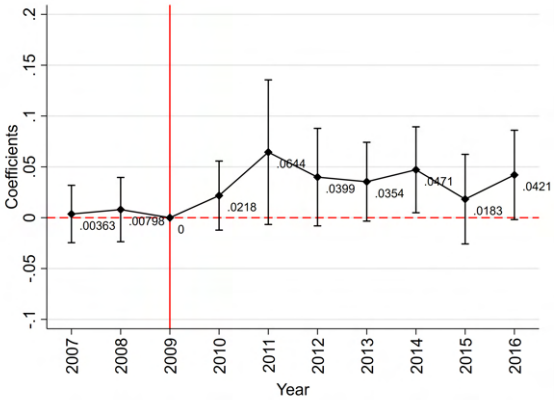
(F) Ave RRB (2002-2005), Replication Stage

Figure A3.4. Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Subjects: Diseases with a Number of Studies ≥ 0

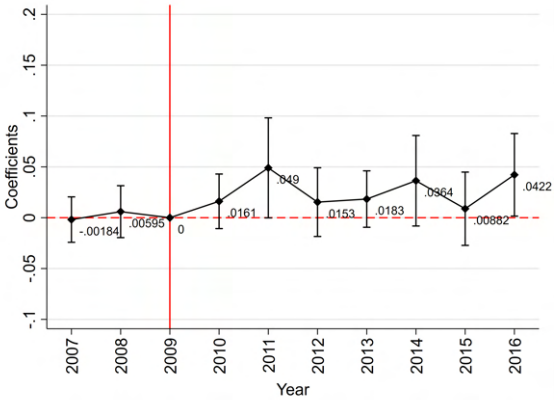
Notes: The sample comprises of 27 broader diseases from 2007 to 2016. There are in total 183 disease-year pairs for analysis combining discovery and replication stages, 182 disease-year pairs for discovery stage analysis, and 170 disease-year pairs for replication stage analysis. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is the share of minority subjects for each disease in each year. The relative referral base (RRB) is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of RRB and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.



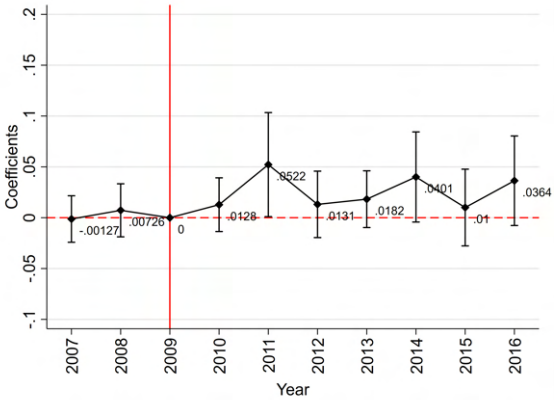
(A) RRB (2005), Discovery + Replication Stages



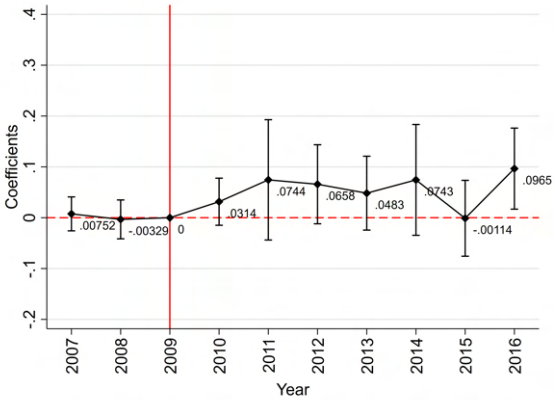
(B) Ave RRB (2002-2005), Discovery + Replication Stages



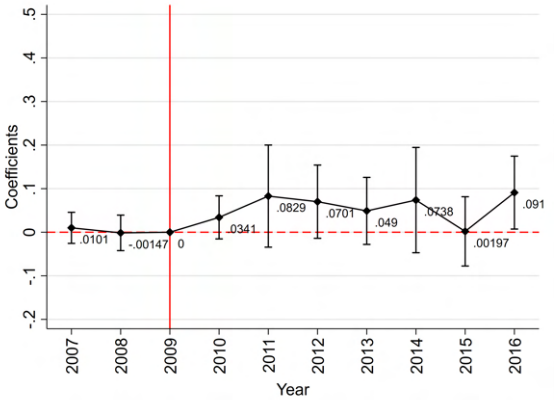
(C) RRB (2005), Discovery Stage



(D) Ave RRB (2002-2005), Discovery Stage



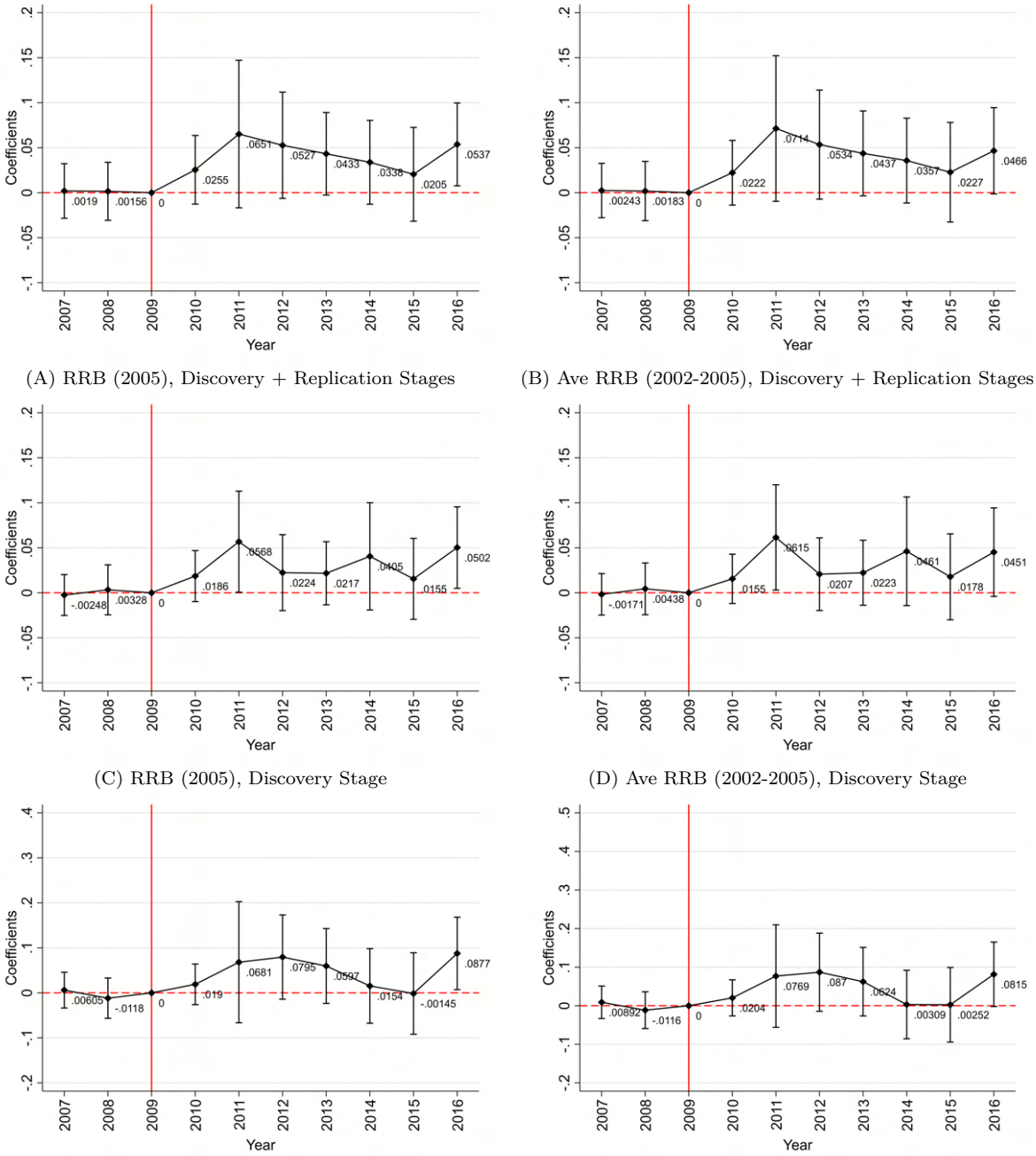
(E) RRB (2005), Replication Stage



(F) Ave RRB (2002-2005), Replication Stage

Figure A3.5. Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Subjects: Diseases with a Number of Studies ≥ 10

Notes: The sample comprises of 27 broader diseases from 2007 to 2016. There are in total 174 disease-year pairs for analysis combining discovery and replication stages, 173 disease-year pairs for discovery stage analysis, and 163 disease-year pairs for replication stage analysis. The sample is constructed based on studies supported by any U.S funding agencies between 2005 and 2016. The dependent variable is the share of minority subjects for each disease in each year. The relative referral base (*RRB*) is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of *RRB* and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.



(A) RRB (2005), Discovery + Replication Stages

(B) Ave RRB (2002-2005), Discovery + Replication Stages

(C) RRB (2005), Discovery Stage

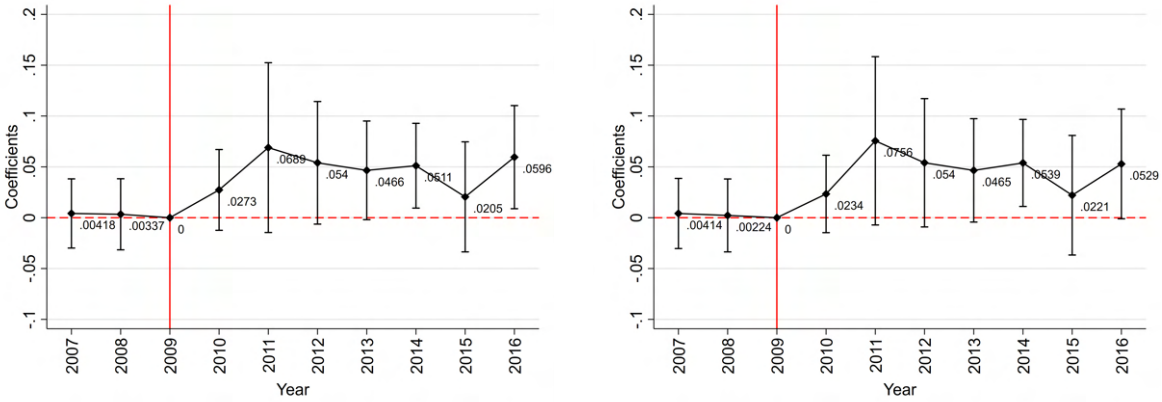
(D) Ave RRB (2002-2005), Discovery Stage

(E) RRB (2005), Replication Stage

(F) Ave RRB (2002-2005), Replication Stage

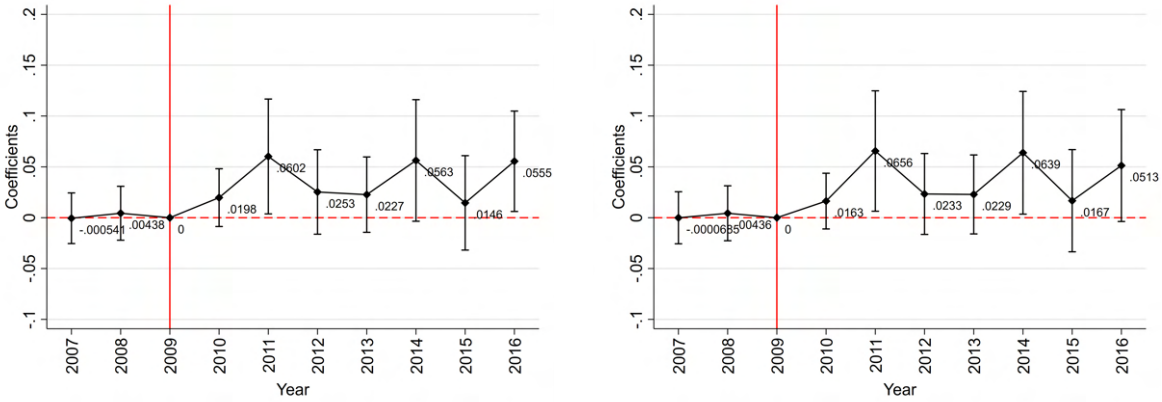
Figure A3.6. Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Subjects: Diseases with a Number of Studies ≥ 20

Notes: The sample comprises of 27 broader diseases from 2007 to 2016. There are in total 157 disease-year pairs for analysis combining discovery and replication stages, 156 disease-year pairs for discovery stage analysis, and 149 disease-year pairs for replication stage analysis. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is the share of minority subjects for each disease in each year. The relative referral base (*RRB*) is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of *RRB* and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.



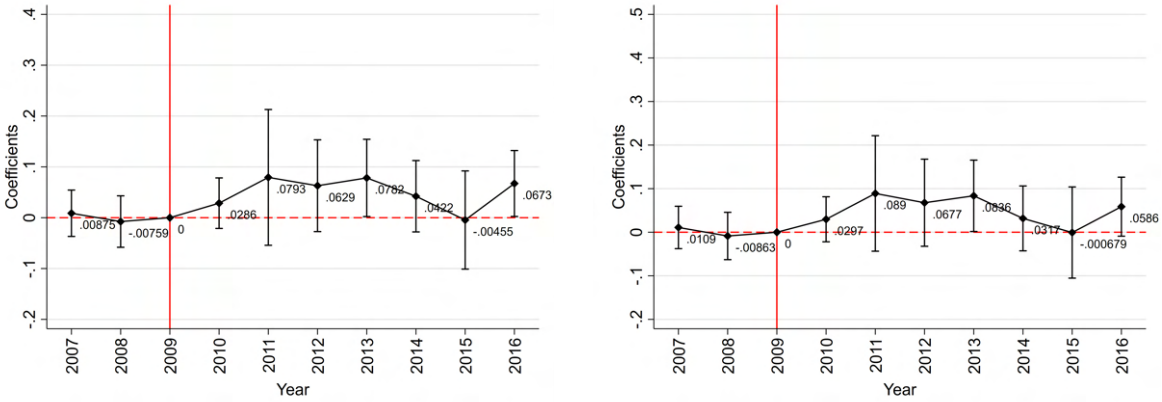
(A) RRB (2005), Discovery + Replication Stages

(B) Ave RRB (2002-2005), Discovery + Replication Stages



(C) RRB (2005), Discovery Stage

(D) Ave RRB (2002-2005), Discovery Stage



(E) RRB (2005), Replication Stage

(F) Ave RRB (2002-2005), Replication Stage

FIGURE A3.7. IMPACT OF ARRA-NIH GRANT EXPANSION ON ANCESTRAL DIVERSITY IN SUBJECTS: DISEASES WITH A NUMBER OF STUDIES ≥ 30

Notes: The sample comprises of 27 broader diseases from 2007 to 2016. There are in total 134 disease-year pairs for analysis combining discovery and replication stages, 133 disease-year pairs for discovery stage analysis, and 127 disease-year pairs for replication stage analysis. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is the share of minority subjects for each disease in each year. The relative referral base (*RRB*) is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of *RRB* and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.

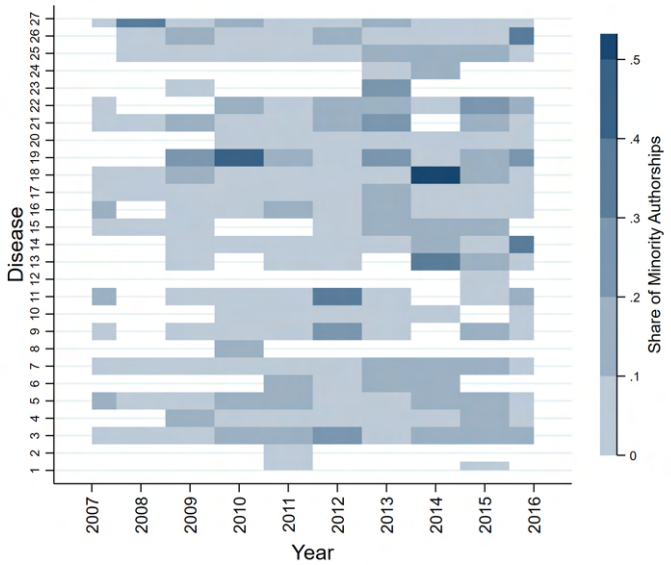
Table A3.1—Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Subjects – Robustness Checks

Sample	Disease with Studies ≥ 0 (1)	Disease with Studies ≥ 10 (2)	Disease with Studies ≥ 20 (3)	Disease with Studies ≥ 30 (4)
Panel A – Key Independent Variable: Relative Referral Base (2005)				
Dependent Variable: Share of Minority Subject				
Relative Referral Base (2005) * Post	0.0341***	0.0345***	0.0409***	0.0434***
	(0.013)	(0.013)	(0.014)	(0.014)
	[0.006]	[0.005]	[0.005]	[0.005]
	[0.0110, 0.0580]	[0.0106, 0.0591]	[0.0189, 0.0628]	[0.0207, 0.0655]
Outcome Mean (Pre-Policy)/Observations	0.051/183	0.051/174	0.051/157	0.051/134
Dependent Variable: Share of Minority Subject in Discovery Sample				
Relative Referral Base (2005) * Post	0.0225**	0.0232**	0.0305**	0.0325***
	(0.010)	(0.010)	(0.012)	(0.012)
	[0.040]	[0.039]	[0.003]	[0.003]
	[0.0011, 0.0440]	[0.0012, 0.0453]	[0.0135, 0.0515]	[0.0153, 0.0541]
Outcome Mean (Pre-Policy)/Observations	0.027/182	0.027/173	0.027/156	0.027/133
Dependent Variable: Share of Minority Subject in Replication Sample				
Relative Referral Base (2005) * Post	0.0485**	0.0490**	0.0465*	0.0493**
	(0.024)	(0.024)	(0.024)	(0.024)
	[0.015]	[0.021]	[0.069]	[0.064]
	[0.0125, 0.0779]	[0.0107, 0.0792]	[-0.0059, 0.0811]	[-0.0041, 0.0853]
Outcome Mean (Pre-Policy)/Observations	0.078/170	0.078/163	0.078/149	0.078/127
Panel B – Key Independent Variable: Average Relative Referral Base (2002-2005)				
Dependent Variable: Share of Minority Subject				
Ave. Relative Referral Base (2002 -2005) * Post	0.0335***	0.0339***	0.0410***	0.0438***
	(0.013)	(0.013)	(0.014)	(0.014)
	[0.008]	[0.006]	[0.006]	[0.004]
	[0.0101, 0.0590]	[0.0096, 0.0601]	[0.0199, 0.0626]	[0.0222, 0.0658]
Outcome Mean (Pre-Policy)/Observations	0.051/183	0.051/174	0.051/157	0.051/134
Dependent Variable: Share of Minority Subject in Discovery Sample				
Ave. Relative Referral Base (2002 -2005) * Post	0.0217**	0.0223**	0.0303**	0.0328***
	(0.010)	(0.010)	(0.012)	(0.012)
	[0.048]	[0.051]	[0.002]	[0.002]
	[0.0002, 0.0437]	[-0.0001, 0.0450]	[0.0138, 0.0493]	[0.0161, 0.0525]
Outcome Mean (Pre-Policy)/Observations	0.027/182	0.027/173	0.027/156	0.027/133
Dependent Variable: Share of Minority Subject in Replication Sample				
Ave. Relative Referral Base (2002 -2005) * Post	0.0496**	0.0500**	0.0478*	0.0510**
	(0.025)	(0.025)	(0.025)	(0.025)
	[0.016]	[0.021]	[0.068]	[0.063]
	[0.0136, 0.0821]	[0.0112, 0.0830]	[-0.0065, 0.0862]	[-0.0056, 0.0910]
Outcome Mean (Pre-Policy)/Observations	0.078/170	0.078/163	0.078/149	0.078/127
Disease Research Field Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Disease FE	Yes	Yes	Yes	Yes

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The relative referral base is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. We report the Huber-White robust standard errors in the parentheses. We also report wild bootstrap cluster p-values in square brackets and wild bootstrap cluster 95% confidence intervals in square brackets, generated using boottest command (10000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at disease level with 27 disease clusters. Significance stars are labeled with respect to the robust standard errors. $*p < 0.1, **p < 0.05, ***p < 0.01$.

A4. Ancestral Diversity in GWAS Authors

(A) Heat Map for the Number of Minority Authors



(B) Heat Map for the Share of Minority Authors

Figure A4.1. Heatmap for Ancestral Diversity in Authors

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. Given minimal GWAS publications in 2005 and 2006, the data in 2007 presented in the figure includes authors from GWAS publications in 2005, 2006 and 2007. We predict the ancestral background associated with each surname based on the 2000 US census. We adopt a conservative rule in prediction of minority authors: a last name is predicted to be associated with a minority background only if its probability of being either "Asian", "Black" or "Hispanic" is more than 0.95. The IDs corresponds to the IDs in Table 1.

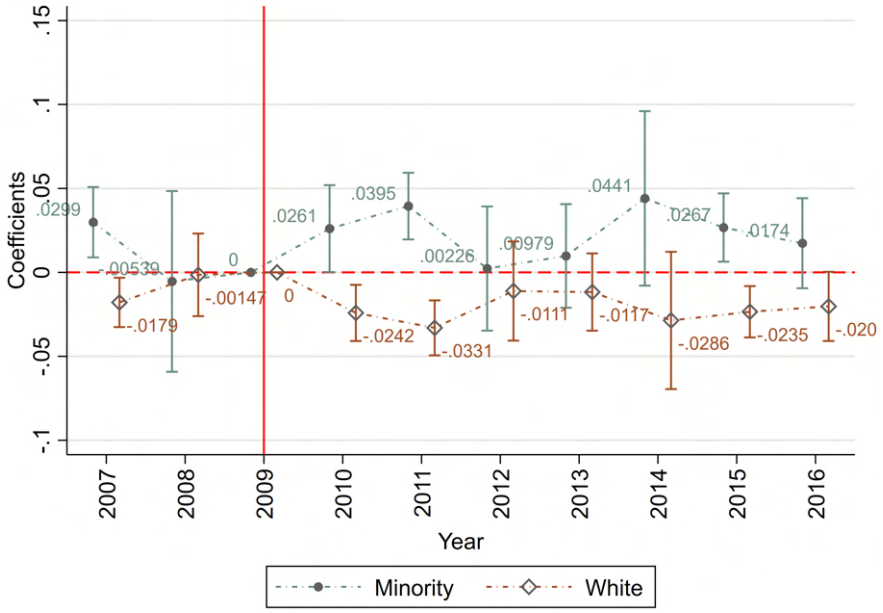
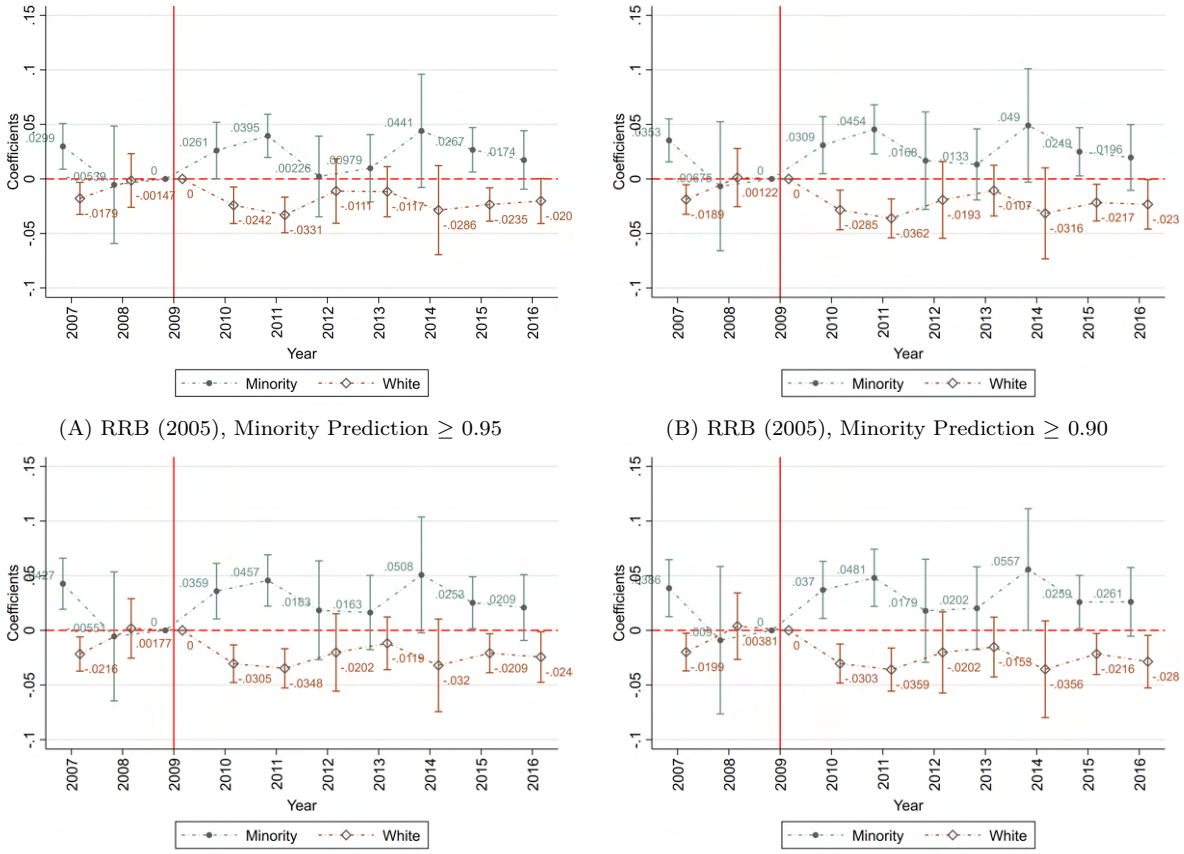


Figure A4.2. Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Authors – RRB Decomposition

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is the share of minority authors for each disease in each year. We adopt a conservative rule in prediction of minority authors: a last name is predicted to be associated with a minority background only if its probability of being either "Asian", "Black" or "Hispanic" is more than 0.95. The referral base (RB) for minorities (whites) is the share of people with medical conditions among minorities (whites). The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of RB and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.



(A) RRB (2005), Minority Prediction $\geq$ 0.95

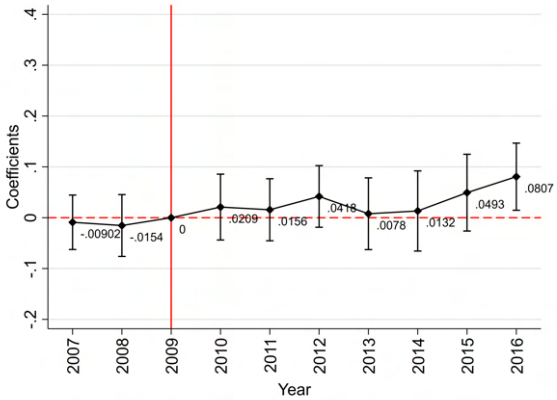
(B) RRB (2005), Minority Prediction $\geq$ 0.90

(C) RRB (2005), Minority Prediction $\geq$ 0.85

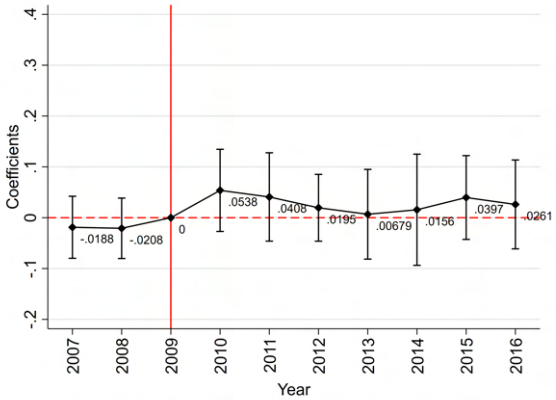
(C) RRB (2005), Minority Prediction $\geq$ 0.8

Figure A4.3. Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Authors – RRB Decomposition, Alternative Criteria

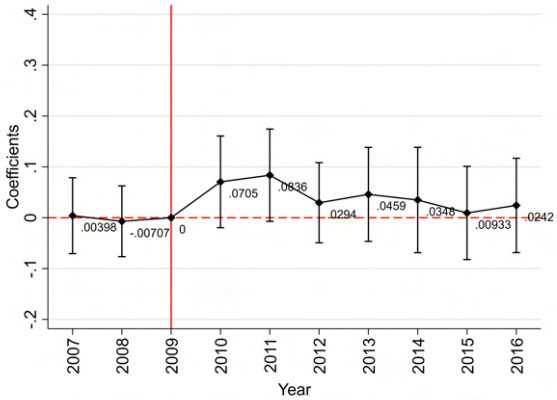
Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is the share of minority authors for each disease in each year. We adopt a different rules in prediction of minority authors: a surname is predicted to be associated with a minority background only if its probability of being either "Asian", "Black" or "Hispanic" is more than 0.95 in Figure (A), 0.90 in Figure (B), 0.85 in Figure (C), and 0.80 in Figure (D). The relative referral base is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of *RRB* and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.



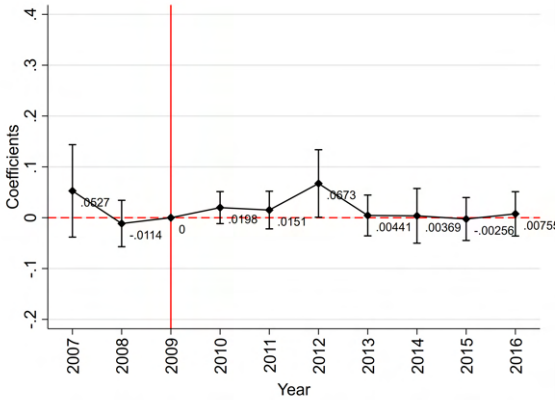
(A) Share of Studies w/ First 1 Author Being Minority



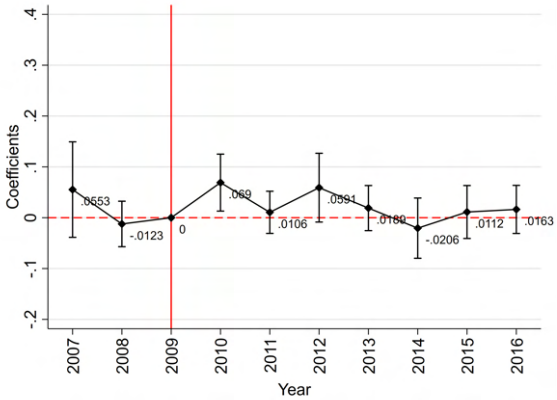
(B) Share of Studies w/ Minority Author in First 2 Authors



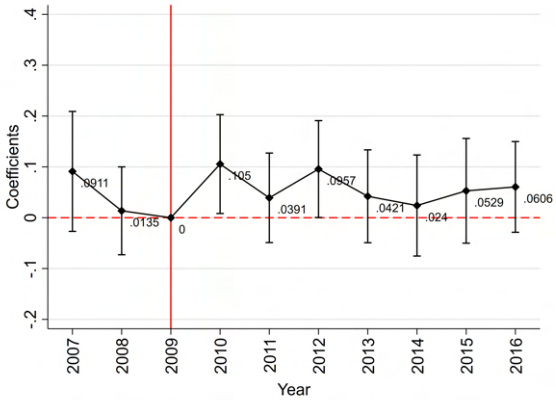
(C) Share of Studies w/ Minority Author in First 3 Authors



(D) Share of Studies w/ Last 1 Author Being Minority



(E) Share of Studies w/ Minority Author in Last 2 Authors



(F) Share of Studies w/ Minority Author in Last 3 Authors

Figure A4.4. Impact of ARRA-NIH Grant Expansion on Collaboration Patterns

Notes: The sample comprises of 27 broader diseases from 2007 to 2016, in total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. The dependent variable is specified under each figure. We adopt a conservative rule in prediction of minority authors: a last name is predicted to be associated with a minority background only if its probability of being either "Asian", "Black" or "Hispanic" is more than 0.95. The relative referral base (RRB) is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. Coefficients of the interactions of RRB and each year indicator are plotted. The 95% confidence intervals based on robust standard errors are plotted.

Table A4.1—Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Authors – Robustness Checks using RRB

Sample	Disease with Studies ≥ 0 (1)	Disease with Studies ≥ 10 (2)	Disease with Studies ≥ 20 (3)	Disease with Studies ≥ 30 (4)
Key Independent Variable: Relative Referral Base (2005)				
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.95				
Relative Referral Base (2005) * Post	0.0145**	0.0130**	0.0119*	0.0118*
	(0.006)	(0.006)	(0.007)	(0.007)
	(0.010)	(0.021)	(0.036)	(0.041)
	[0.0052, 0.0230]	[0.0030, 0.0210]	[0.0012, 0.0198]	[0.0011, 0.0190]
Outcome Mean (Pre-Policy)/Observations	0.075/183	0.075/174	0.075/157	0.075/134
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.9				
Relative Referral Base (2005) * Post	0.0168**	0.0155**	0.0145**	0.0145**
	(0.007)	(0.007)	(0.007)	(0.007)
	(0.010)	(0.017)	(0.031)	(0.034)
	[0.0060, 0.0257]	[0.0039, 0.0241]	[0.0022, 0.0230]	[0.0019, 0.0223]
Outcome Mean (Pre-Policy)/Observations	0.094/183	0.094/174	0.094/157	0.094/134
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.85				
Relative Referral Base (2005) * Post	0.0162**	0.0150**	0.0141*	0.0143*
	(0.007)	(0.007)	(0.008)	(0.008)
	(0.014)	(0.023)	(0.040)	(0.041)
	[0.0048, 0.0255]	[0.0029, 0.0239]	[0.0011, 0.0234]	[0.0011, 0.0228]
Outcome Mean (Pre-Policy)/Observations	0.105/183	0.105/174	0.105/157	0.105/134
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.8				
Relative Referral Base (2005) * Post	0.0177**	0.0166**	0.0149*	0.0152*
	(0.007)	(0.007)	(0.008)	(0.008)
	(0.009)	(0.018)	(0.032)	(0.035)
	[0.0062, 0.0270]	[0.0044, 0.0260]	[0.0018, 0.0239]	[0.0017, 0.0236]
Outcome Mean (Pre-Policy)/Observations	0.114/183	0.114/174	0.114/157	0.114/134
Disease Research Field Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Disease FE	Yes	Yes	Yes	Yes

Notes: The most complete sample comprises of 27 broader diseases from 2007 to 2016, with a total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. We also require the disease research domain to have a minimum number of studies as specified in each column. The dependent variable is the share of minority authors for each disease in each year. We adopt different rules in prediction of minority authors: a surname is predicted to be associated with a minority background only if its probability of being either "Asian", "Black" or "Hispanic" is more than 0.95, 0.90, 0.85 or 0.80. The relative referral base is the difference in the share of people with medical conditions between minorities and whites. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. We report the Huber-White robust standard errors in the parentheses. We also report wild bootstrap cluster p-values in square brackets and wild bootstrap cluster 95% confidence intervals in square brackets, generated using boottest command (10000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at disease level with 27 disease clusters. Significance stars are labeled with respect to the robust standard errors. $*p < 0.1, **p < 0.05, ***p < 0.01$.

Table A4.2—Impact of ARRA-NIH Grant Expansion on Ancestral Diversity in Authors – Robustness Checks using 2002-2005 Ave. RRB

Sample	Disease with Studies ≥ 0 (1)	Disease with Studies ≥ 10 (2)	Disease with Studies ≥ 20 (3)	Disease with Studies ≥ 30 (4)
Key Independent Variable: Average Relative Referral Base (2002-2005)				
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.95				
Ave. Relative Referral Base (2002 -2005) * Post	0.0138**	0.0123*	0.0109	0.0109
	(0.006)	(0.006)	(0.007)	(0.007)
	(0.016)	(0.034)	(0.067)	(0.068)
	[0.0038, 0.0220]	[0.0014, 0.0202]	[-0.0012, 0.0182]	[-0.0014, 0.0175]
Outcome Mean (Pre-Policy)/Observations	0.075/183	0.075/174	0.075/157	0.075/134
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.9				
Ave. Relative Referral Base (2002 -2005) * Post	0.0161**	0.0149**	0.0137*	0.0136*
	(0.007)	(0.007)	(0.008)	(0.008)
	(0.014)	(0.026)	(0.048)	(0.052)
	[0.0046, 0.0246]	[0.0025, 0.0231]	[0.0002, 0.0212]	[-0.0002, 0.0205]
Outcome Mean (Pre-Policy)/Observations	0.094/183	0.094/174	0.094/157	0.094/134
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.85				
Ave. Relative Referral Base (2002 -2005) * Post	0.0157**	0.0145**	0.0134	0.0134
	(0.007)	(0.007)	(0.008)	(0.008)
	(0.021)	(0.034)	(0.061)	(0.063)
	[0.0035, 0.0245]	[0.0015, 0.0230]	[-0.0009, 0.0219]	[-0.0012, 0.0212]
Outcome Mean (Pre-Policy)/Observations	0.105/183	0.105/174	0.105/157	0.105/134
Dependent Variable: Share of Minority Authors – Prob. Cutoff of 0.8				
Ave. Relative Referral Base (2002 -2005) * Post	0.0170**	0.0159**	0.0139	0.0138
	(0.008)	(0.008)	(0.008)	(0.009)
	(0.016)	(0.029)	(0.058)	(0.072)
	[0.0045, 0.0259]	[0.0024, 0.0249]	[-0.0008, 0.0221]	[-0.0019, 0.0217]
Outcome Mean (Pre-Policy)/Observations	0.114/183	0.114/174	0.114/157	0.114/134
Disease Research Field Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Disease FE	Yes	Yes	Yes	Yes

Notes: The most complete sample comprises of 27 broader diseases from 2007 to 2016, with a total 183 disease-year pairs. The sample is constructed based on studies supported by any US funding agencies between 2005 and 2016. We also require the disease research domain to have a minimum number of studies as specified in each column. The dependent variable is the share of minority authors for each disease in each year. We adopt different rules in prediction of minority authors: a surname is predicted to be associated with a minority background only if its probability of being either "Asian", "Black" or "Hispanic" is more than 0.95, 0.90, 0.85 or 0.80. The 2002-2005 average relative referral base is the average difference in the share of people with medical conditions between minorities and whites between 2002 and 2005. The disease research field characteristics includes the total number of studies funded by only non-US grants up to the previous year, the number of minority-based SNPs discovered in studies funded by only non-US grants up to the previous year, and the total number of grants from the US funding agencies up to the previous year. We report the Huber-White robust standard errors in the parentheses. We also report wild bootstrap cluster p-values in square brackets and wild bootstrap cluster 95% confidence intervals in square brackets, generated using boottest command (10000 replications) in Stata 14 (Roodman et al. 2019) for standard errors clustered at disease level with 27 disease clusters. Significance stars are labeled with respect to the robust standard errors. $*p < 0.1, **p < 0.05, ***p < 0.01$.

Appendix B is available upon request.